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Workpackage WPB

Implementation – Online Job Vacancies

Technical report on the implementation requirements of prototypes in the relevant statistical production processes at European and national level

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1 Contents

2 Summary 5

3 OJA implementation requirements 6

4 OJA Specific Application services developed in the frame of the project 9

4.1 Acquisition and recording 9

4.1.1 Specific scraping 9

4.1.1.1 Domain Scraper 9

4.1.1.2 Enterprise Scraper 9

4.1.1.3 Romanian websites Scrapers 9

4.2 Data wrangling 9

4.2.1 Pseudo-stock transformation 9

4.3 Data representation 10

4.3.1 Data classification 10

4.3.1.1 Job Title Processing 10

4.4 Modeling and interpretation 10

4.4.1 Data linking & enriching 10

4.4.1.1 Nowcasting JVS 10

4.4.1.2 Forecasting economic indicators from OJA 10

4.4.2 Statistical aggregation 10

4.4.2.1 Labour market concentration-index 10

4.4.2.2 AnalysisOutput 11

4.4.2.3 Online Job Advertisements experimental statistics 11

4.4.2.4 Automated monthly vacancy report 11

5 References 13

List of Figures:

Figure 1: Example of BREAL and OJA specific application services..... 8

List of Tables:

Table 1: BREAL and OJA specific application services 7

2 Summary

The aims of implementing this work package were to produce statistical estimates on the topic of online job adverts and to identify statistical production processes and capabilities that may be affected at the national level and to define the conceptual production processes at the national level and at the level of the ESS. Suitable techniques and specific methodologies were developed during the pilot phase of the project ESSNet Big Data 2016-2018 on Web Scraping for Job Vacancy Statistics. The implementation phase was based on work carried out to establish the conditions under which web scraping techniques can be used and to evaluate the quality of the scraped data.

Using online job adverts data (OJA) has a number of advantages. They can be used successfully for the needs of official statistics for completing and estimating the number and characteristics of the job vacancies. Their main advantages are their accessibility, timeliness, comparatively low cost of producing them and lack of or minimal burden for the respondents. Most OJA contain information not only for the vacant position but also for the required qualifications and skills, including the so-called soft skills, for degree and kind of the graduate education and professional experience necessary for the potential applicants for taking it. However, using OJA as an alternative information source for statistical surveys is always accompanied by challenges.

Since European Centre for the Development of Vocational Training (CEDEFOP) is developing a pan-European system for collecting and processing of OJA data, ESSNet collaborated with CEDEFOP on using their data with the focus on facilitating the integration of these data in the future Web Intelligence Hub (WIH). WIH will enable NSIs to focus more on statistical challenges and further methodological development.

The final aim of the activities was to develop and test the methodology and prototypes as well as capacity building to facilitate their integration into production at the level of individual NSI and at the level of the ESS.

The aim of this document is to describe developed prototypes for collecting, processing and analysis of OJA data. The first part, OJA implementation requirements, outline the architecture and processes for supporting OJA data integration and production of statistical outputs from OJA. The second part, prototypes and methods for processing OJA data, describes developed prototypes and their implementation requirements.

3 OJA implementation requirements

The current methods in official statistics are based on specific features of traditional statistical data, small amounts of high quality data. These data are collected to fit a specific and predefined statistical purpose and are usually well structured. They are usually collected to ensure a given coverage of population, definitions, methodology and quality in order to meet the needs of official statistics. The guarantee that statistical data are of good quality is the responsibility of the domain expert who designed the survey. This scenario changes when dealing with OJA data. These data can have any kind of structure, they may be structured or semi-structured (data collected from job portals) or completely unstructured (companies' web pages). These data are scraped continuously from the internet, with dynamic changes of webpages and job portals.

The quality of data depends on ability in collecting and pre-processing the data (identifying duplicated adverts, etc.), linked to statistical data (registers), as well as combining with other statistical data sources. There is a need for developing new models and tools to support the integration of OJA data sources into official statistics. Tools and prototypes developed in the frame of the project are described in the next chapter.

Various approaches for collecting and processing OJA data uses similar data processing steps. However they brings important considerations for implementation architecture particularly in the case of common OJA productions system. It is important to define where the used data (OJA data, JVS data, register data) are located (local, European or worldwide level) and how the services (data collection, data processing, information extraction) are autonomous, interoperable, replicated or shared between partners. An application architecture is a map of how an organization's software applications are assembled as part of its overarching enterprise architecture and how those applications interact with each other to meet business or user requirements.

In the case of OJA data processing steps BREAL Big Data Reference Architecture and Layers - Business Layers [1] defines respective business functions:

- Acquisition and Recording
- Data Wrangling
- Data Representation
- Modelling and Interpretation
- Shape output
- Validation

According to BREAL each business function consists of several composing functions. For some of them, it is relevant to identify some Application Services, i.e. logical components belonging to a software application layer that implement the specific functions. Application services in Table 1 are described in two levels:

- BREAL Generic Application services
- External Application services

BREAL Generic Application services implements business functions independently from domain of interest whilst External application services implements the same business functions but in OJA

specific and detailed way. This approach also support the addition of domain specific application services when identified, for example Pseudo-stock transformation service as shown in Figure 1 with the business functions in yellow and the application services in blue.

Table 1: BREAL and OJA specific application services

BREAL Business function	BREAL Generic Application services	OJA Specific Application services
Acquisition and recording	External structured data retrieval	Specific scraping, API access
	External unstructured data retrieval	Generic scraping, Website retrieval
	Data storing	Document storing, Relational storing, Key-value storing,
Data wrangling	Data preparation, filtering and deduplication	Data filtering, Language detection, Deduplication and merging
	Data standardization	Format standardisation, Text processing, Data representation and enrichment
	Data aggregation	Pseudo-stock transformation
Data representation	Data derivation	Compute new variables, Entity recognition
	Structure modelling	Data classification
	Data encoding	
Modeling and interpretation	Data linking & enriching	Deduplication by linking
	Methodology application	
	Statistical aggregation	
Shape output	Data exchange	

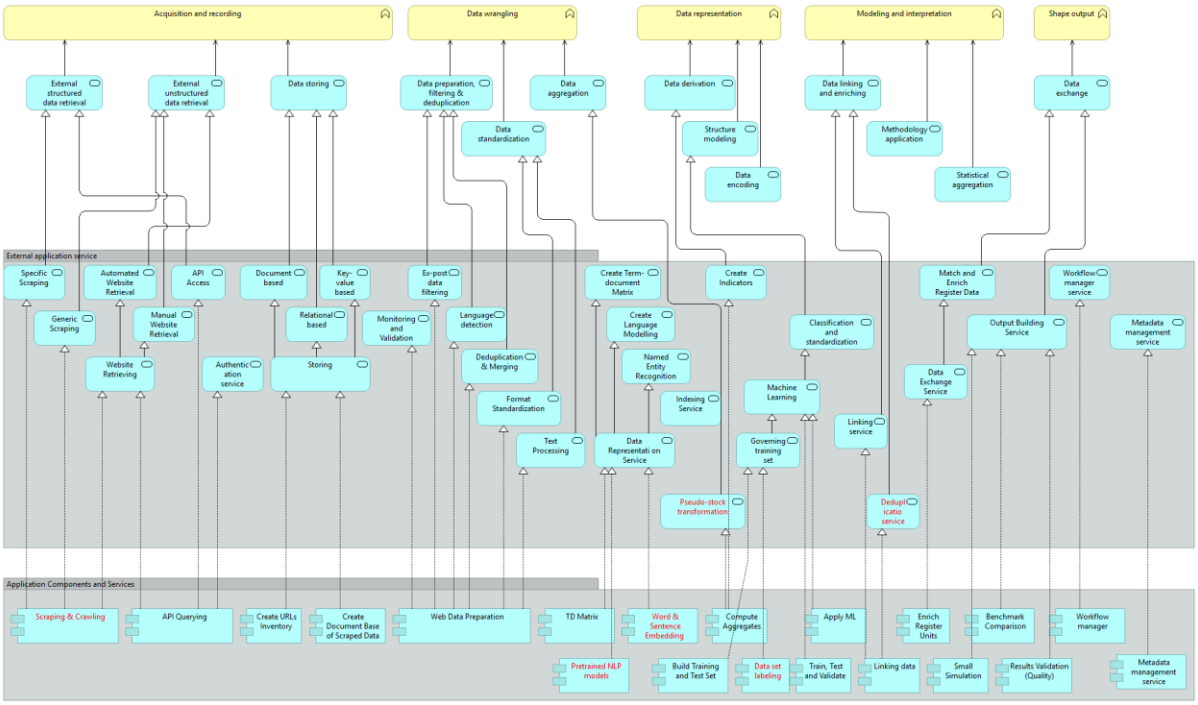

Figure 1: Example of BREAL and OJA specific application services

4 OJA Specific Application services developed in the frame of the project

The following chapters describes developed prototypes and their implementation requirements according to defined Application services. Detailed description and implementation requirements can be found on GitHub https://github.com/OnlineJobVacanciesESSnetBigData.

4.1 Acquisition and recording

4.1.1 Specific scraping

4.1.1.1 Domain Scraper

General programmes and documentation for scraping of an individual domain and its subpages. The Scraper takes the name of the project (and the resulting files, be they to be saved), the target internet domain or a list of target pages and other keyword arguments (are described in detail in the relevant document) as input. Before starting the scraping process, the robot will check the (first) target page's robots.txt and Robots.txt page for scraping restrictions. Users can also specify Firefox profiles to run the Scraper with and it will in turn use the same profile on the Driver objects.

Detailed description and implementation requirements can be found on GitHub https://github.com/OnlineJobVacanciesESSnetBigData/SURS_DomainScraper.

4.1.1.2 Enterprise Scraper

General programmes and documentation for scraping of enterprise pages that were deemed potential job advertisements by the crawling process. The job of this spider is to collect the text contents on all sites and save it into a file that is used for analysis and processing. The spider has 5 specific methods defined according to the standard scrapy guidelines.

Detailed prototypes description and implementation requirements can be found on project GitHub https://github.com/OnlineJobVacanciesESSnetBigData/SURS_EnterpriseScraper.

4.1.1.3 Romanian websites Scrapers

Programmes for scrapping data from some Romanian websites. Detailed description and implementation requirements can be found on GitHub https://github.com/OnlineJobVacanciesESSnetBigData/romScrappers.

4.2 Data wrangling

4.2.1 Pseudo-stock transformation

The data layout and descriptive material from CEDEFOP hint at an unstructured approach to collecting OJA-data by scraping new ads whenever there happen to be resources available. This results in large day-to-day fluctuations in quantity, coverage and choice of source which we observe in the data. It also means that we cannot interpret the raw data as flow or stock. In order to address this issue, we explore the possibility of generating pseudo-stocks from CEDEFOP data which are an approximation to the underlying stock of job ads. R code for calculating and plotting monthly and quarterly pseudo-stocks of job ads from the CEDEFOP data for Germany

This code is intended to be run on the EU Dataplatform and can be found on GitHub https://github.com/OnlineJobVacanciesESSnetBigData/Pseudostocks-from-CEDEFOP-OJA.

4.3 Data representation

4.3.1 Data classification

4.3.1.1 Job Title Processing

A package to clean job titles and classify them into occupation nomenclature. It was designed for French language but French resources are separated from the main code. It should be fairly easy to add your own language. Detailed description and implementation requirements can be found on GitHub https://github.com/OnlineJobVacanciesESSnetBigData/JobTitleProcessing_FR

4.4 Modeling and interpretation

4.4.1 Data linking & enriching

4.4.1.1 Nowcasting JVS

A Jupyter notebook file that demos how nowcasting can be done. Also included are two mock files - one is representing (mock) job vacancy survey (JVS) data time-series, the other mock online job vacancy data time series. The latter was derived from the JVS using random perturbations on the data (of different scale for each industry), including situations when the way OJV approximates JVS suddenly changes. The Jupyter notebook is commented and can be simply opened and run. If you wish to try it on your own datasets, simply modify the paths/replace the mock files with your own, preserving the format of the data. Detailed description and implementation requirements can be found on GitHub https://github.com/OnlineJobVacanciesESSnetBigData/jv-polished

4.4.1.2 Forecasting economic indicators from OJA

An example of usage of OJA to forecast an economic indicator used in Slovenia as a component for the calculation of the GDP. This Notebook presents an example of usage of OJA data to try and estimate 2 time series that are used in the Slovenian calculation of the GDP. The target data consists of 38 monthly values for the VAT of Turnover from the sale of services and the Index of Turnover from the sale of services. Detailed description and implementation requirements on GitHub https://github.com/OnlineJobVacanciesESSnetBigData/SURS_EconomicIndicatorFromOJA.

4.4.2 Statistical aggregation

4.4.2.1 Labour market concentration-index

R code used to calculate a labour market concentration index as in Azar et al. (2018), with German CEDEFOP data. The index measures how concentrated a local labour market is from the perspective of a potential employee. For each occupation, region and time period, a limited number of potential employers are posting vacancies. Depending on the number and market shares of employers, this local labour market segment can be very concentrated. This concept is similar to market power born of concentration in product markets. This might allow employers to force workers to accept wages below equilibrium levels, because moving or waiting is costly for employees.

The index implemented here uses the following definitions for local labour market segments:

- Occupation by isco_level 4
- Timeframe of one quarter
- Geographic region in terms of functional urban areas (EU definition)

As input, it uses job ad data from CEDEFOP, provided by the EU Dataplatform. Detailed description and implementation requirements can be found on GitHub

https://github.com/OnlineJobVacanciesESSnetBigData/Labour-market-concentration-index-from-CEDEFOP-data.

4.4.2.2 AnalysisOutput

This repository contains some programmes for data analysis against CEDEFOP and national datasets. First file contains code for fitting discrete distributions for national and datasets. Second file is built against the DataLab and generates plots for several variables from the CEDEFOP dataset. The results are contained by a number of graphs related to number of online job ads and a break down by different categories of variables (data collection period, economic sector, educational level, esco_level_1 and experience requirements). Detailed description and implementation requirements can be found on GitHub https://github.com/OnlineJobVacanciesESSnetBigData/romAnalysisOutput.

4.4.2.3 Online Job Advertisements experimental statistics

Bulgarian National Statistical Institute software used for Online Job Advertisements experimental statistics. It scrapes data from web sites described in JSON files in format by webscraper.io Google Chrome extension. It is used for collecting data about OJA, accommodations and online prices in e-shops. The software process the scraped data every day and generates weekly, monthly and quarterly reports with experimental statistical data. Detailed description and implementation requirements can be found on GitHub https://github.com/OnlineJobVacanciesESSnetBigData/BNSI-OJA-software.

4.4.2.4 Automated monthly vacancy report

R markdown document, which illustrates what a monthly job vacancy report could look like. It is closely related to the monthly vacancy reports published by the Australian government. It contains:

- the (pseudo-)stock of job ads for the current month and its change relative to the previous month and the same month last year
- the monthly time series of job ads, superimposed on the time series of unemployment
- stock and annual change of job ads by German federal state
- stock and annual change of job ads by occupation group (ISCO level 1)
- stock and annual change of job ads by eduction level
- top 10 most sought occupations current month (ISCO level 3)
- top 10 companies which advertise the most jobs in the current month
- geographic distribution of job ads in Germany by functional urban area

As input, it uses job ad data from CEDEFOP, provided by the EU Dataplatform.Detailed description and implementation requirements can be found on GitHub

https://github.com/OnlineJobVacanciesESSnetBigData/Automated_monthly_vacancy_report_DE.

5 References

[1] BREAL Big data reference architecture and Layers - Business Layer, Available at: https://webgate.ec.europa.eu/playground-multisite/crosportal/content/WPF_Milestones_and_deliverables_en